

DevOps - 3 Month Complete Course Syllabus (Approx. 90 Hours)

Week 1-2: Introduction to DevOps & Linux Basics (10 hours)

- What is DevOps and Why DevOps?
- DevOps Lifecycle & Tools Overview
- Linux Basics: Shell, Commands, File System
- Permissions, Users, and Package Management

Week 3: Version Control with Git & GitHub (8 hours)

- Git Basics: Init, Clone, Commit, Push, Pull
- Branching, Merging, Rebase
- Working with GitHub: Repositories, Pull Requests
- Git Workflows: Git Flow, Forking

Week 4: Build Tools & CI with Jenkins (10 hours)

- Introduction to CI/CD Concepts
- Installing and Configuring Jenkins
- Creating and Managing Jenkins Jobs
- Pipelines: Scripted vs Declarative
- Integrating Git with Jenkins

Week 5: Configuration Management with Ansible (8 hours)

- What is Ansible and Why use it?
- Ansible Architecture and Installation
- Writing and Executing Playbooks
- Variables, Handlers, Roles
- Inventory and Ad-hoc Commands

Week 6-7: Docker - Containerization (12 hours)

- What is Docker & Containerization?
- Installing Docker, Docker CLI Basics
- Creating Dockerfiles and Building Images
- Docker Volumes, Networks

- Docker Compose for Multi-Container Apps

Week 8-9: Kubernetes - Orchestration (12 hours)

- Introduction to Kubernetes & Architecture
- Minikube Setup and kubectl Usage
- Pods, Deployments, ReplicaSets
- Services, ConfigMaps, and Secrets
- Helm Basics (Optional)

Week 10: Monitoring & Logging (8 hours)

- Importance of Monitoring & Logging
- Prometheus for Metrics Collection
- Grafana for Visualization
- Log Management with ELK Stack (Intro)

Week 11: Cloud Platforms & Infrastructure as Code (10 hours)

- Cloud Overview: AWS/GCP/Azure
- Deploying a Web App on Cloud
- Infrastructure as Code: Introduction to Terraform
- Provisioning Infrastructure with Terraform

Week 12: Final Capstone Project & DevOps Best Practices (12 hours)

- Designing End-to-End CI/CD Pipeline
- Containerized App Deployment on Kubernetes
- Monitoring Setup with Prometheus/Grafana
- Backup & Rollback Strategies
- DevOps Best Practices & Interview Prep